

Data Collection & Preparation

Date	2 October 2026
Team ID	SWTID-2026-6582
Project Name	PocketSmart AI: Your Smart Budget & Recommendation Assistant
Maximum Marks	Project Deliverable

Data Source Identification

- Direct user inputs from Home, Party and Jewelry planner forms.
- Optional user-uploaded outfit image for Jewelry Planner.
- Google Gemini API output returned as structured recommendation data.
- SQLite records created for authenticated user accounts and recommendation history.

Data Collection Method

Data is collected at request time through HTML forms. The project does not rely on a pre-trained custom dataset or a scraped marketplace dataset. Marketplace/service navigation is generated as search URLs; prices shown by the application are AI-generated estimates.

Data Cleaning & Preprocessing

- Pydantic validates numeric ranges and required fields.
- Email, username and password rules are validated.
- Images are restricted to supported formats, maximum 5 MB, and verified using Pillow.
- Planner inputs are formatted into structured prompts for Gemini.
- Gemini structured output is parsed into typed schemas before financial validation.

Data Validation & Quality Check

Budget must be positive and within configured limits; guest count is constrained; room/item counts have upper bounds; malformed images are rejected. AI-generated items are recalculated by Python to verify total spending.

Data Privacy & Ethical Considerations

API keys and application secrets are kept in environment variables. Passwords are stored only as bcrypt hashes. Uploaded images are used for the requested styling analysis. The application should not represent estimated prices as verified live merchant prices.

Input-Output Examples

Input	Example Output
Party: ₹85,000, 50 guests, Birthday, Chennai, selected services	Structured categories/items with estimated unit prices, quantities, explanations, total estimated spending and remaining budget.
Jewelry: budget + occasion + optional outfit image	Outfit analysis (when image supplied) plus coordinated jewelry recommendations and validated budget totals.